

# ABHILASH NEOG

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## Research Summary

Doctoral researcher working on foundation models, with research spanning complex sequential and spatiotemporal data, multimodal learning and generative modeling. My work includes foundation models for irregular and sparse time-series, flow-based generative models, and vision-language models for evaluation, grounding, and scientific applications. I am currently focused on cross-modal alignment and generative models for image generation/editing.

## Education

### Virginia Tech

Ph.D., Computer Science. Advisor: *Dr. Anuj Karpatne*. GPA: 4.0/4.0

Aug 2022 - May 2027

Blacksburg, USA

### Virginia Tech

M.S., Computer Science. Advisor: *Dr. Anuj Karpatne*. GPA: 4.0/4.0

Aug 2022 - Dec 2024

Blacksburg, USA

### Birla Institute of Technology and Science (BITS), Pilani

Bachelor of Engineering (B.E.), Computer Science, GPA: 8.08/10

July 2016 – July 2020

Pilani, India

## Work Experience

### Applied Scientist Intern | Microsoft AI

May 2026 – Aug 2026

- Proposed a novel training framework for flow-based generative models (FLUX) – via adaptive multiscale wavelet supervision, trained via LoRA fine-tuning – for preserving subject clarity in image outpainting. *Resulted in a research paper (under review)*
- Developed a novel image outpainting pipeline via (a) VLM-based semantic guidance, (b) a high-frequency patch estimation f/w, & (c) an image injection technique for additional conditioning in FLUX models. *Resulted in a patent submission (under review)*

### Machine Learning Intern | Kryptowire

May 2023 – Aug 2023

- Developed an outlier detection model for denoising sensor-based Human Activity Recognition (HAR) **time series** data
- Built & deployed a CNN-based HAR model achieving 82% F-1 score on an android app using Keras & TensorFlow Lite

### Data Scientist | Oracle

Sep 2020 – July 2022

- Built & deployed Machine Learning applications into ETL pipelines, leveraging Spark systems, MLOps & CI/CD pipelines
- Designed and deployed a *Demand Prediction* application for time series forecasting using the DeepAR model
- Developed an unsupervised classification algorithm achieving 40% higher accuracy than then LMs on a 71k-label dataset.

### Software Engineer Intern | VMware

Jan 2020 – June 2020

- Streamlined the process of fetching & filtering raw data from Workspace ONE Cloud using Spring Boot REST APIs
- Contributed to an end-user federation app on Workspace ONE Cloud, and wrote unit tests using JUnit and Mockito

### Software Engineer Intern | Samsung Research Institute

May 2019 – July 2019

- Performed a feasibility study of Multi-frame Noise Reduction solutions' deployment in Live Focus for Low light conditions
- Optimized the existing HAL call flow, in C++, with considerable noise reduction in the first phase of live focus capture

## Research Experience

### Graduate Research Assistant | Knowledge-guided Machine Learning Lab

Jan 2023 – Present

- Working on cross-modal alignment (LLMs and time-series foundation models) for Q/A, and grounded prediction tasks
- Contributed to a novel physics-guided Diffusion Model for solving Partial Differential Eqns., achieving *15-250x* faster inference
- Zero-shot evaluation of Vision-Language Models (VLMs) for *scene graph generation* and *reasoning ability* on VQA tasks involving scientific images
- Proposed a Model-agnostic Adaptation technique for Irregular and Sparse Time-Series (TS) modeling
- Proposed an (a) *effective tokenization strategy* and (b) *Query-based forecasting*, to train a Foundation Model for spatio-temporal (probabilistic) predictions on irregular grids, leveraging multi-gpu training.

## Talks/Awards

- Invited talk at the *ASLO 2026 Conference - Session on AI for the Next Frontier in Aquatic Sciences*
- Invited talk at the *AGU 2025 Conference - Session on Advancing Water Science Through AI*
- Talk on Transfer Learning in Lake Ecosystems at the “*NSF Macrosystems Biology Meeting 2024*”
- Lightning Talk at the “*Frontiers in Ecological Forecasting 2023*,” event at Virginia Tech
- NSF NAIRR (National AI Research Resource) Pilot Award, 2024*
- “*Star of the Month (Dec 2021)*” within the Oracle Analytics Cloud Organization, Oracle India

## Publications

1. **Abhilash Neog**, Taewan Kim, Yi Wu, Xu Chen, Jian Jiao. “Preserving Subject-Clarity in Image Outpainting with Multiscale Wavelet Supervision”. *Under Review*
2. **Abhilash Neog**, Sepideh Fatemi, Medha Sawhney, et al. “LakeFM: Toward a Foundation Model for Aquatic Ecosystems Using Irregular Multivariate Multi-depth Time Series Data”. *KDD 2026* [pdf]
3. **Abhilash Neog**, Arka Daw, Sepideh Fatemi, Medha Sawhney, et al. “Investigating a Model-Agnostic and Imputation-Free Approach for Irregularly-Sampled Multivariate Time-Series Modeling”. *TMLR, 2026* [pdf]
4. A. Pandey, **Abhilash Neog**, Gautam Jajoo. “On the Internal Semantics of Time-Series Foundation Models”. *BERT2S @ NeurIPS 2025* [pdf]
5. M. Sawhney, **Abhilash Neog**, Mridul Khurana, Anuj Karpatne. “PRISMA: Improving the Accuracy–Latency Frontier of Diffusion-based PDE Solvers Using Physics-Informed Spectral Attention”. *Under review (ML4PS @ NeurIPS 2025)* [pdf]
6. S. Fatemi, **Abhilash Neog**, Amartya Dutta, M. Sawhney, et al. “Scientific Equation Discovery using Modular Symbolic Regression via Vision-Language Guidance”. *CV4Science @ CVPR 2025 (Oral)*
7. A. Dutta, M. Sawhney, K.S. Mehrab, **Abhilash Neog**, Mridul Khurana, et al. “Open World Scene Graph Generation using Vision Language Models”. *World Models @ ICML 2025* [pdf]
8. KS Mehrab, M. Maruf, Arka Daw, **Abhilash Neog**, HB Manogaran, et al. “Fish-Vista: A Multi-Purpose Dataset for Understanding & Identification of Traits from Images”. *CVPR 2025* [pdf]
9. M. Maruf, Arka Daw, KS Mehrab, HB Manogaran, **Abhilash Neog**, M. Sawhney, et al. “VLM4Bio: A Benchmark Dataset to Evaluate Pretrained Vision-Language Models for Trait Discovery from Biological Images”. *NeurIPS 2024* [pdf]
10. Lavika Goel, **Abhilash Neog**, Ashish Aman, and Arshveer Kaur. “Hybrid Nature-Inspired Optimization Techniques in Face Recognition” *Transactions on Computational Science XXXVI, Springer LNCS, 2020*. [pdf]

## Patents

1. **Abhilash Neog**, Taewan Kim, Yi Wu, Xu Chen. *Microsoft, 2026*. “Subject-Clarity-Aware Image Outpainting via Semantic and Structural Guidance with Vision-Language Models and High-Frequency Map Estimation.” *Patent submission (under review)*.
2. Akash B., Krishnan R., **Abhilash, N.**, Dipawesh P. and Karthik B. *Oracle, 2024*. “Machine Learning Based Spend Classification.” *U.S. Patent Application 17/903,161*. [pdf]

## Selected Projects

- Can Large Vision Language Models (VLMs) Ground Fine-grained Attribute?**  [pdf](#) **Aug '24 – Dec '24**
- Developed a novel dual-scale attention framework for fine-grained attribute localization in Large Vision-Language Models (LVLMs), incorporating entropy-based head selection, maximally connected component filtering, and hierarchical constraints
- Evaluating Model Reasoning & Hallucinations in Medical LLMs**  [code](#)  [pdf](#) **Jan '24 – April '24**
- Analyzed and evaluated factual error propagation in open-source medical LLMs such as BioMistral, Asclepius, Alpacare, and PMC-LLaMA to identify variations in their efficacy and ensure reliable information dissemination in medical settings.
- Mathematical Reasoning in Large Language Models (LLMs)**  [code](#)  [pdf](#) **Aug '23 – Dec '23**
- Worked on the problem of numerical headline generation and numeral masked-fill as part of NumEval @ SemEval 2024
  - Adapted **Llama, T5, BART & RoBERTa** models by Direct **fine-tuning & prompt engineering** for the respective tasks
- Text Summarization of Electronic Theses and Dissertations (ETD)**  [pdf](#) **Sept '22 – Dec '22**
- Developed a text summarization pipeline, integrating both Transformer-based abstractive algorithms (pre-trained Pegasus & RoBERTa) and traditional extractive algorithms like TextRank, LexRank & LSA, within an ETD Info. Retrieval system

## Academic Duties

- **Reviewing:**
  - Conferences/Journal - NeurIPS ('25, '26), AAI ('26, '27), TMLR, ICML 2026, KDD 2026, ECCV 2026
  - Workshops - FM4Science (ICLR '26), FMSD (ICML '25, '26), FMs in the Wild (ICLR '25), Open Science for FMs (ICLR '25), XAI4Science (ICLR '25), ML4PS (NeurIPS '25), TSALM (NeurIPS '24)
- **Teaching:**
  - Graduate Teaching Assistant, CS 5805 Machine Learning, Spring 2024
  - Teaching Assistant, BITS F312 Neural Networks and Fuzzy Logic, Fall 2019

## Technical Skills

**Languages:** Python, Java, C++, SQL, R

**Frameworks:** PyTorch, Git, Hydra, Tensorflow Keras, Spark